



SUMMARY

AI & HPC young engineer with a computer vision background (MSc specialized in AI & Computer Vision), experienced in transformers, **large-scale distributed training** and **efficient inference**. Interested in **computer vision and AI for science**; **biological imaging** is especially motivating to me as a challenging application domain. Seeking a full-time ML role in Paris, available from September 2026 (flexible).

WORK EXPERIENCE

Research Analyst Intern – Flatiron Institute / Simons Foundation October 2025 – May 2026 New York, NY

- **Independently designed and deployed an end-to-end platform for scientific foundation model inference and web visualization** on Kubernetes, integrating multiple services (frontend, gateway, media processing, inference) to showcase the company's new scientific model performance online.
- Developed and benchmarked **transformer training and inference** across multiple GPU architectures, informing hardware selection and optimization strategies.
- Conducted performance analysis of **low-precision quantized inference** which has been presented at **DP2E-AI** : International workshop on distributing large-scale AI systems.

Assistant Software Engineer Intern – IDRIS / CNRS France August 2023 – January 2024 Paris, France

- Implemented and benchmarked **distributed deep-learning strategies** (DDP, FSDP) on a large-scale multi-node GPU cluster, supporting researchers in training large models.

SKILLS

- Machine Learning & Deep Learning: **PyTorch, Transformers, Distributed Training**, vLLM, TensorRT-LLM, Quantization, Hyperparameter Optimization, **Image Segmentation**, Data Augmentation
- Performance & Distributed Systems: Nvidia systems (Cuda Toolkit) and architectures, large-scale ML
- Programming: **Python**, C/C++, Bash, Git, Web (React, JS, CSS)
- Systems & Infrastructure: Linux, Slurm, Docker, Kubernetes

SELECTED PROJECTS

- **Polymathic AI** — Inference server + web interface for multi-domain scientific foundation models (biology, astronomy). (*React, Kubernetes*)
- **Rock Segmentation** — CV pipeline to segment and count sand-grain-sized particles using watershed and morphological ops. (OpenCV, scikit-image)
- **Deep RL agents** (Trackmania, Snake, Pac-Man) — Vision-based agents trained end-to-end in PyTorch.
- **Self-hosted LLM on Kubernetes** — Scalable inference on local GPUs.

EDUCATION

Dual-degree program:

- University of Technology of Belfort-Montbéliard (UTBM) 2020 – September 2026* Belfort, France
Master's degree in engineering in Computer Science, specialized in **AI & Computer Vision**.
- University Of Quebec at Chicoutimi (UQAC) 2024 – September 2026* Chicoutimi, Canada
Master's degree in Computer Science, specialized in **AI**.

*Expected – Coursework completed